



SIKSHA 'O' ANUSANDHAN

(Deemed to be University)

Faculty of Engineering & Technology (ITER)

Department of Computer Science and Engineering

Project Proposal Form

FINAL YEAR RESEARCH PROJECT-2026

SECTION: 29

GROUP NO: 13

PROJECT TITLE: Proactive Carbon Intelligence: ML-Based Emission Forecasting and Anomaly Detection

PROJECT ABSTRACT:

Industrial facilities are among the largest contributors to global greenhouse gas emissions, primarily due to inefficient energy utilization and the absence of predictive sustainability analytics. Most industries rely on retrospective reporting systems that identify excessive energy consumption and emission spikes only after they occur, leading to financial losses and environmental damage. The aim of this project, *Verdify: A Predictive AI Framework for Industrial Energy Waste and Emission Optimization*, is to develop an intelligent, data-driven system capable of forecasting energy consumption, detecting anomalous emission patterns, and recommending proactive optimization strategies.

The proposed system utilizes structured industrial datasets containing historical energy usage, CO₂ emission levels, operational load parameters, and environmental variables. Advanced machine learning techniques are employed, including Long Short-Term Memory (LSTM) networks for time-series forecasting, Isolation Forest and Autoencoder models for anomaly detection, and regression-based modelling for efficiency optimization. The framework continuously analyses temporal patterns to predict potential emission spikes and energy waste before they occur.

A sustainability analytics dashboard visualizes forecasts, anomaly alerts, and carbon reduction simulations, enabling decision-makers to take preventive action. The system is evaluated using performance metrics such as MAE, RMSE, and F1-score to ensure predictive reliability and robustness.

The primary benefit of this project lies in shifting industrial sustainability from a reactive to a proactive paradigm. By providing early warning signals and optimization insights, Verdify can reduce operational costs, lower carbon footprints, and support data-backed ESG compliance. This research contributes to the growing field of AI-driven environmental intelligence and demonstrates how predictive analytics can accelerate the transition toward sustainable industrial ecosystems.

(1) SOFTWARE, HARDWARE OR METHODS/ALGORITHMS SPECIFICATIONS:

A. Software Specifications

Programming Language:

- Python 3.x

Development Environment:

- Jupyter Notebook / VS Code

Libraries & Frameworks:

- NumPy, Pandas (Data preprocessing)
- Matplotlib, Seaborn, Plotly (Visualization)
- Scikit-learn (Machine Learning models)
- TensorFlow / PyTorch (Deep Learning models – LSTM, Autoencoders)
- Statsmodels (ARIMA/SARIMA time-series modelling)
- Flask / Streamlit (Dashboard deployment)

Database (Optional):

- SQLite / PostgreSQL for structured data storage

Operating System:

- Windows / Linux / macOS

B. Hardware Specifications

Since no physical IoT deployment is included, only computational resources are required:

- Processor: Intel i5 / Ryzen 5 or higher
- RAM: Minimum 8 GB (16 GB recommended for deep learning training)
- Storage: 256 GB SSD or higher
- GPU (Optional but recommended): NVIDIA GPU with CUDA support for faster LSTM training

C. Methods / Algorithms Specifications

The system consists of three major AI modules:

1. Time-Series Forecasting

Purpose: Predict future energy consumption and CO₂ emission levels.

Algorithms:

- LSTM (Long Short-Term Memory Networks)
- GRU (Gated Recurrent Units)
- ARIMA / SARIMA (Baseline statistical models)

Evaluation Metrics:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Square Error)

- MAPE (Mean Absolute Percentage Error)

2. Anomaly Detection

Purpose: Detect abnormal energy spikes or unusual emission patterns.

Algorithms:

- Isolation Forest
- One-Class SVM
- Deep Autoencoders

Evaluation Metrics:

- Precision
- Recall
- F1-Score

3. Optimization Modelling

Purpose: Estimate optimal energy usage and simulate emission reduction.

Methods:

- Multiple Linear Regression
- Gradient Boosting Regressor
- Reinforcement Learning-based policy optimization (optional advanced module)

(2) NAME, REG. NO AND SIGNATURE OF GROUP MEMBERS:

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(3) APPROVAL STATUS (To be filled in by the Section Coordinator of FRP):

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Project Supervisor

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Section Coordinator, FRP

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FRP Coordinator